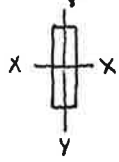


APPLICATIONS OF THE AREA MOMENT OF INERTIA (I)

I. The Area Moment of Inertia (I) integral $I_x = \int_A y^2 dA$ arises frequently in Mechanics of Materials. For this reason, values of this integral are tabulated for commonly used shapes (see other side for examples). One should also be able to calculate I for simple shapes via integration or the Parallel Axis Thm. $I = \bar{I} + Ad^2$.

II. The examples below will use a 2"x6" unfinished white pine board.



$$I_{xx} = \frac{1}{12}(2)(6^3) = 36 \text{ in}^4$$

$$I_{yy} = \frac{1}{12}(6)(2^3) = 4 \text{ in}^4$$

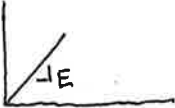
If this board was planed to 1 1/2" x 5 1/2" :

$$I_{xx} = \frac{1}{12}(1\frac{1}{2})(5.5^3) = 20.8 \text{ in}^4$$

$$I_{yy} = \frac{1}{12}(5.5)(1.5^3) = 1.55 \text{ in}^4$$

To apply the examples, also need E (Elastic Modulus or Young's Modulus) for White Pine

Stress
 σ
psi



$$\text{Strain, } \frac{\text{in}}{\text{in}} = \frac{\Delta L}{L}$$

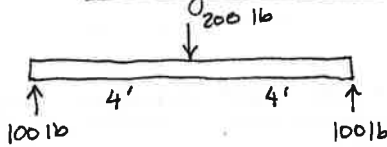
E for white pine = $1.6 \times 10^6 \text{ psi}$

To compare: $E_{\text{STEEL}} = 30 \times 10^6 \text{ psi (MILD)}$

This is akin to Hooke's Law for a spring.

It is the resistance of a material to deformation when a load is applied

III. Bending Stress and Deflection : Consider 8 ft length of 2x6. 200 lb load at center. Knowledge of statics gives us the shear^(V) and bending moment^(M) values for this beam (See Beam Diagrams below)



V & M depend only on length and loading -- not on material or shape of beam.

Max Bending Moment: $M = \frac{PL}{4} = \frac{(200)(8)}{4} = 400 \text{ lb ft}$

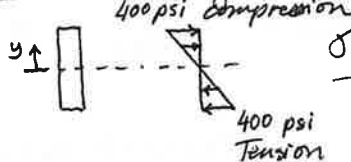
Bending Stress

$$\sigma = \frac{My}{I}$$

Deflection (Max)

$$\Delta_{\text{max}} = \frac{PL^3}{48EI}$$

A. Board Standing on Smaller End (Planed Board)



$$\sigma_{\text{max}} = \frac{(400 \text{ lb ft}) \left(\frac{5\frac{1}{2}}{2} \right) \left(\frac{12 \text{ in}}{\text{ft}} \right)}{20.8 \text{ in}^4} = 635 \text{ psi}$$

Zero stress at centroid of area.

This is called the neutral axis.

Deflection: $\Delta_{\text{max}} = \frac{(200) \left(\frac{8 \text{ ft}}{2} \right)^3}{48(1.6 \times 10^6 \text{ psi})(20.8 \text{ in}^4)} \left(\frac{144 \text{ in}^2}{\text{ft}^2} \right) = 0.00923 \text{ ft}$
 $= 0.1108 \text{ in}$
 $\approx \frac{1}{8} \text{ in}$

B. Board Laying Flat



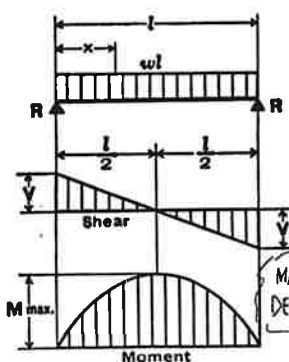
$$\sigma_{\text{max}} = \frac{(400 \text{ lb ft}) \left(\frac{1\frac{1}{2}}{2} \right) \left(\frac{12 \text{ in}}{\text{ft}} \right)}{1.55 \text{ in}^4} = 2320 \text{ psi}$$

Deflection: $\Delta_{\text{max}} = \frac{(200 \text{ lb})(8 \text{ ft})^3}{48(1.6 \times 10^6 \text{ psi})(1.55 \text{ in}^4)} \left(\frac{144 \text{ in}^2}{\text{ft}^2} \right) = 0.124 \text{ ft}$
 $\approx 1.5 \text{ in}$

BEAM DIAGRAM SHEETS AND FORMULAS For various static loading conditions

AMERICAN INSTITUTE OF STEEL CONSTRUCTION

1. SIMPLE BEAM—UNIFORMLY DISTRIBUTED LOAD



Total Equiv. Uniform Load = wl

$R = V \dots \dots \dots = \frac{wl}{2}$

$V_x \dots \dots \dots = w \left(\frac{l}{2} - x \right)$

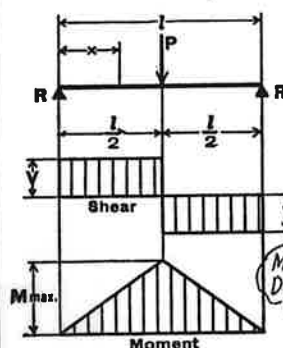
$M_{\text{max. (at center)}} \dots = \frac{wl^2}{8}$

$M_x \dots \dots \dots = \frac{wx}{2} (l - x)$

$\Delta_{\text{max. (at center)}} \dots = \frac{5wl^4}{384EI}$

$\Delta_x \dots \dots \dots = \frac{wx}{24EI} (l^3 - 2lx^2 + x^3)$

7. SIMPLE BEAM—CONCENTRATED LOAD AT CENTER



Total Equiv. Uniform Load = $2P$

$R = V \dots \dots \dots = \frac{P}{2}$

$M_{\text{max. (at point of load)}} \dots = \frac{Pl}{4}$

$M_x \text{ (when } x < \frac{l}{2}) \dots = \frac{Px}{2}$

$\Delta_{\text{max. (at point of load)}} \dots = \frac{Pl^3}{48EI}$

$\Delta_x \text{ (when } x < \frac{l}{2}) \dots = \frac{Px}{48EI} (3l^2 - 4x^2)$

ZZ axis is the axis of minimum moment of inertia for these shapes.

ANGLES
Equal legs and unequal legs
Properties for designing

Location of centroid + Location of Axis

Size and Thickness	k	Weight per Ft	Area	Axis X-X					
				I	S	r	y	Z	y _p
In.	In.	Lb.	In. ²	In. ⁴	In. ³	In.	In.	In. ³	In.
L 6x6 x1	1 1/2	37.4	11.0	35.5	8.57	1.80	1.86	15.5	0.917
3/4	1 1/8	33.1	9.73	31.9	7.63	1.81	1.82	13.8	0.811
3/4	1 1/4	28.7	8.44	28.2	6.66	1.83	1.78	12.0	0.703
3/4	1 1/2	24.2	7.11	24.2	5.66	1.84	1.73	10.2	0.592
3/4	1 3/4	21.9	6.43	22.1	5.14	1.85	1.71	9.26	0.536
3/4	2	19.6	5.75	19.9	4.61	1.86	1.68	8.31	0.479
3/4	2 1/4	17.2	5.06	17.7	4.08	1.87	1.66	7.34	0.422
3/4	2 1/2	14.9	4.36	15.4	3.53	1.88	1.64	6.35	0.363
3/4	2 3/4	12.4	3.65	13.0	2.97	1.89	1.62	5.35	0.304
L 6x4 x 3/4	1 1/8	27.2	7.98	27.7	7.15	1.86	2.12	12.7	1.44
3/4	1 1/4	23.6	6.94	24.5	6.25	1.88	2.08	11.2	1.38
3/4	1 1/2	20.0	5.86	21.1	5.31	1.90	2.03	9.51	1.31
3/4	1 3/4	18.1	5.31	19.3	4.83	1.90	2.01	8.66	1.28
3/4	2	16.2	4.75	17.4	4.33	1.91	1.99	7.78	1.25
3/4	2 1/4	14.3	4.18	15.5	3.83	1.92	1.96	6.88	1.22
3/4	2 1/2	12.3	3.61	13.5	3.32	1.93	1.94	5.97	1.19
3/4	2 3/4	10.3	3.03	11.4	2.79	1.94	1.92	5.03	1.16

What is the actual I_{zz} ?

From radius of gyration, $r = \sqrt{\frac{I_{zz}}{A}} = 1.17$

$I_{zz} = 1.17^2 (11.0) = 15.1 \text{ in}^4$
(for 6x6x1 Angle)

ANGLES
Equal legs and unequal legs
Properties for designing

For Plastic Design

Size and Thickness	Axis Y-Y						Axis Z-Z	
	I	S	r	x	Z	x _p	r	Tan α
In.	In. ⁴	In. ³	In.	In.	In. ³	In.	In.	α
L 6x6 x1	35.5	8.57	1.80	1.86	15.5	0.917	1.17	1.000
3/4	31.9	7.63	1.81	1.82	13.8	0.811	1.17	1.000
3/4	28.2	6.66	1.83	1.78	12.0	0.703	1.17	1.000
3/4	24.2	5.66	1.84	1.73	10.2	0.592	1.18	1.000
3/4	22.1	5.14	1.85	1.71	9.26	0.536	1.18	1.000
3/4	19.9	4.61	1.86	1.68	8.31	0.479	1.18	1.000
3/4	17.7	4.08	1.87	1.66	7.34	0.422	1.19	1.000
3/4	15.4	3.53	1.88	1.64	6.35	0.363	1.19	1.000
3/4	13.0	2.97	1.89	1.62	5.35	0.304	1.20	1.000
L 6x4 x 3/4	9.75	3.39	1.11	1.12	6.31	0.665	0.857	0.421
3/4	8.68	2.97	1.12	1.08	5.47	0.578	0.860	0.428
3/4	7.52	2.54	1.13	1.03	4.62	0.488	0.864	0.435
3/4	6.91	2.31	1.14	1.01	4.19	0.442	0.866	0.438
3/4	6.27	2.08	1.15	0.987	3.75	0.396	0.870	0.440
3/4	5.60	1.85	1.16	0.964	3.30	0.349	0.873	0.443
3/4	4.90	1.60	1.17	0.941	2.85	0.301	0.877	0.446
3/4	4.18	1.35	1.17	0.918	2.40	0.252	0.882	0.448

IV. Column Buckling - Consider a column w/ simply supported ends (not fixed or clamped)

Max load before buckling:

$$P_{CRIT} = \frac{\pi^2 EI}{L^2}$$

I is the minimum moment of inertia for the column:

A. Consider the 2x6" pine:

$$P_{CRIT} = \frac{\pi^2 (1.6 \times 10^6) (4 \text{ in}^4)}{(8 \text{ ft})^2 (144 \text{ in}^2/\text{ft}^2)} = 6850 \text{ lb}$$

B. Planed Board:

$$P_{CRIT} = \frac{\pi^2 (1.6 \times 10^6) (1.547 \text{ in}^4)}{(8 \text{ ft})^2 (144)} = 2650 \text{ lb}$$

C. 6x6x1 Steel Angle Shown Above. $I_{zz} = r^2 A = (1.17^2) (11.0) = 15.1 \text{ in}^4$

$$P_{CRIT} = \frac{\pi^2 (30 \times 10^6) (15.1)}{(8 \text{ ft})^2 (144)} = 485 \text{ kip}$$

Nominal Depth of Beam
↓
Designation
↓
Wt. per foot

"W" ≡ Wide Flanged Beams
Sometimes will see :
"S SHAPES" for
"Standard Beams"

W SHAPES
Dimensions

Designation	Area A	Actual Depth d	Web		Flange		Distance					
			Thickness t_w	$\frac{t_w}{2}$	Width b_f	Thickness t_f	T	k	k_1			
	In. ²	In.	In.	In.	In.	In.	In.	In.	In.			
W 8x 67	19.7	9.00	9	0.570	$\frac{3}{8}$	8.280	8 $\frac{1}{4}$	0.935	$\frac{1}{8}$	6 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
x 58	17.1	8.75	8 $\frac{1}{2}$	0.510	$\frac{1}{2}$	8.220	8 $\frac{1}{4}$	0.810	$\frac{1}{8}$	6 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
x 48	14.1	8.50	8 $\frac{1}{2}$	0.400	$\frac{3}{8}$	8.110	8 $\frac{1}{4}$	0.885	$\frac{1}{8}$	6 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
x 40	11.7	8.25	8 $\frac{1}{4}$	0.360	$\frac{3}{8}$	8.070	8 $\frac{1}{4}$	0.560	$\frac{3}{8}$	6 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
x 35	10.3	8.12	8 $\frac{1}{4}$	0.310	$\frac{3}{8}$	8.020	8	0.495	$\frac{1}{2}$	6 $\frac{1}{2}$	1	$\frac{1}{8}$
x 31	9.13	8.00	8	0.285	$\frac{3}{8}$	7.995	8	0.435	$\frac{3}{8}$	6 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
W 8x 28	8.25	8.06	8	0.285	$\frac{3}{8}$	6.535	6 $\frac{1}{2}$	0.465	$\frac{3}{8}$	6 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
x 24	7.08	7.93	7 $\frac{7}{8}$	0.245	$\frac{1}{4}$	6.495	6 $\frac{1}{2}$	0.400	$\frac{3}{8}$	6 $\frac{1}{2}$	$\frac{7}{8}$	$\frac{1}{8}$

(Elastic) Section Modulus: $S = \frac{I}{y}$
e.g. $S_{xx} = \frac{272}{(9/2)} = 60.4 \text{ in}^3$
y = dist. from centroidal axis to extreme edge of beam

W SHAPES Properties

Radius of Gyration $r = \sqrt{\frac{I}{A}} = \sqrt{\frac{272}{19.7}} = 3.72 \text{ in}$

Nominal Wt. per Ft	Compact Section Criteria		F _y	X ₁	X ₂ × 10 ⁶	Elastic Properties						Plastic Modulus	
	b _f /2t _f	h _w /t _w				Axis X-X			Axis Y-Y			Z _x	Z _y
						I	S	r	I	S	r		
Lb.			Ksi	Ksi	(1/Ksi) ²	In. ⁴	In. ³	In.	In. ⁴	In. ³	In.	In. ³	In. ³
67	4.4	11.1	—	6620	73.9	272	60.4	3.72	88.6	21.4	2.12	70.2	32.7
58	5.1	12.4	—	5820	122	228	52.0	3.65	75.1	18.3	2.10	59.8	27.9
48	5.9	15.8	—	4860	238	184	43.3	3.61	60.9	15.0	2.08	49.0	22.9
40	7.2	17.6	—	4080	474	146	35.5	3.53	49.1	12.2	2.04	39.8	18.5
35	8.1	20.4	—	3610	761	127	31.2	3.51	42.6	10.6	2.03	34.7	16.1
31	9.2	22.2	—	3230	1180	110	27.5	3.47	37.1	9.27	2.02	30.4	14.1
28	7.0	22.2	—	3480	931	98.0	24.3	3.45	21.7	6.63	1.62	27.2	10.1
24	8.1	25.8	—	3020	1610	82.8	20.9	3.42	18.3	5.63	1.61	23.2	8.57